Banking Transaction Analytics

Project Report: Power BI Dashboard for Transaction, Customer and Financial Performance Analysis

Prepared by: **Sarthak Gedam** | Date: October 2026 | Tools: Excel, Python, MySQL, Power BI | Data source: MySQL (Banking_Transactions database)

1. Executive Summary

This project delivers an interactive three-page Power BI dashboard that analyses 550,000 retail and business banking transactions made by 79,916 unique customers between January 2019 and January 2024. The dashboard gives management a single view of transaction volumes and values, payment success and fraud, customer distribution, and loan and merchant activity, with slicers for account type, KYC status, loan category and loan type. The work followed a three-stage workflow: Excel and Python for initial data analysis, MySQL queries to solve specific business problems, and Power BI to present the results in an easy-to-understand report so that businesses can make better decisions.

Headline results

- The bank processed **550,000 transactions worth ₹16.45 billion**, with an average ticket of ₹29,907 but a median of only ₹2,036, so a small number of very large payments drives total value.
- **92.04%** of transactions succeeded; 3.98% failed, 2.00% were reversed and 1.97% remain pending.
- **4,873 transactions (0.886%)** were flagged as fraud, worth ₹582.1 million (3.54% of total value).
- **RTGS is the main concentration of value and risk:** it is 6.0% of transactions but 54.0% of value, and has a fraud rate of 1.85%, about twice the portfolio average.
- **UPI leads on volume** (28.0% of transactions) but contributes only 1.3% of value.
- **12.1% of transactions** come from customers whose KYC status is Pending or Expired, which is a compliance exposure of about ₹2.03 billion.

2. Project Background and Objectives

Banks generate large volumes of transaction data, but raw tables do not show where growth, risk and customer value sit. This project turns a transaction-level dataset into insight in three stages: initial analysis in Excel and Python, business problem solving with MySQL queries, and a Power BI dashboard that makes the results easy to understand and supports better business decisions.

Objectives

1. Build a clean, reusable data model and a small set of standard KPIs (volume, value, success rate, fraud rate, customer counts).
2. Analyse transaction behaviour by type, direction, year and KYC status.
3. Profile customers by channel, state and account type.
4. Assess financial performance by account type, merchant category and loan category.
5. Highlight fraud and operational risk so that follow-up actions can be prioritised.

3. Data Description

Source. The data is stored in a MySQL database (Banking_Transactions, table bank). It was first explored in Excel and Python, queried in MySQL, and then loaded into Power BI using Power Query. The table is named banking_transactions bank in the model.

Scope. 550,000 rows and 23 columns, one row per transaction. Transaction IDs are unique and no missing values were found in any column. The date range is 1 January 2019 to 1 January 2024.

Group	Fields
Transaction	Transaction_Id, Transaction_Date, Transaction_Time, Transaction_Hour, Year, Month_Name, Transaction_Type, Transaction_Amount, Transaction_Direction, Transaction_Status, Channel, Merchant_Category, Is_Fraud
Customer and account	Customer_Id, Account_Type, Account_Balance, State, Credit_Score, Kyc_Status
Loan	Has_Loan, Loan_Type, Loan_Category, Emi_Amount

Dimensions in the data

- 5 account types, 10 transaction types, 6 channels, 15 merchant categories, 10 states.
- 4 transaction statuses: Success, Failed, Reversed, Pending.
- 3 KYC statuses: Verified, Pending, Expired.
- 7 loan types and 8 loan categories (including No Loan and Unknown Loan Type).

4. Methodology

1. **Initial analysis (Excel and Python).** Explored the dataset in Excel and Python to understand its structure, check data quality and review distributions before deeper analysis.
2. **Business problem solving (MySQL).** Stored the data in MySQL and wrote SQL queries to answer business questions on transaction volumes and values, payment success, fraud, customer segments and loans.
3. **Data import.** Connected Power BI to the MySQL database and imported the bank table using Power Query.
4. **Data preparation and modelling.** Set data types for dates, amounts and identifiers, and prepared Year, Month_Name and Loan_Category fields (six named loan categories plus No Loan and Unknown Loan Type). Used a single fact table with automatic date hierarchies, so no joins were required.
5. **Measures.** Wrote eight DAX measures (Section 5) so that every visual uses one consistent definition of each KPI.
6. **Visualisation.** Designed three report pages (1920 × 1060) with a shared KPI strip, four dropdown slicers and a consistent colour theme, so that non-technical users can explore the data easily.
7. **Validation.** Cross-checked the headline KPIs against the underlying table to confirm that they match the dashboard measures.

5. Data Model and DAX Measures

Measure	DAX logic	Purpose
Total_Transactions	DISTINCTCOUNT of Transaction_Id	Transaction volume
Total_Trans_Amt	SUM of Transaction_Amount	Transaction value
Successful_Transactions	Total_Transactions where Transaction_Status = "Success"	Completed payments
Success_Rate	DIVIDE(Successful_Transactions, Total_Transactions, 0)	Payment reliability
Fraud_Transactions	Total_Transactions where Is_Fraud = 1	Fraud cases
Fraud_Rate	DIVIDE(Fraud_Transactions, Total_Transactions, 0)	Fraud exposure
Total_customers	COUNT of Customer_Id	Customer records (counts rows)
Unique_Customers	DISTINCTCOUNT of Customer_Id	Distinct customers

DIVIDE is used for both rates so that empty slicer selections return 0 instead of an error.

6. Dashboard Design

Each page has the same header, a KPI strip of six cards (Total Transactions, Total Transaction Amount, Successful Transactions, Fraud Transactions, Success Rate, Fraud Rate) and four slicers: Account Type, KYC Status, Loan Category and Loan Type. Selecting a slicer value filters every visual on the page.

Page	Visuals
Transaction Trends	Transaction amount by transaction type (bar); transactions by KYC status (donut); transaction amount by direction, Debit vs Credit (pie); transactions by year (line); transaction amount by year (column)
Customer Insights	Customers by channel (bar); total vs unique customers (scatter); unique customers by state (column); customers by account type (pie)
Financial Performance	Transactions by account type (bar); transactions by merchant category (column); customers by loan category (bar); transactions by month (line)

7. Key Findings

7.1 Overall performance

KPI	Value
Total transactions	550,000
Total transaction amount	₹16.45 billion
Unique customers	79,916
Average transactions per customer	6.9 (median 7, maximum 20)
Successful transactions	506,247 (92.04%)
Fraud transactions	4,873 (0.886%)

7.2 Transaction trends

Volume and value are driven by different payment types.

Transaction type	Share of transactions	Share of value	Average amount (₹)	Fraud rate
UPI	28.0%	1.3%	1,359	0.79%
IMPS	14.0%	3.3%	7,020	0.81%
NEFT	12.1%	16.8%	41,464	0.87%
POS	11.0%	0.7%	1,996	0.80%
ATM withdrawal	10.0%	0.8%	2,424	0.83%
Net banking	7.0%	5.0%	21,317	0.85%
RTGS	6.0%	54.0%	269,254	1.85%
Auto debit	5.0%	0.5%	2,704	0.68%
Cheque	3.9%	17.2%	130,362	1.18%
Credit card	3.0%	0.5%	5,047	0.85%

- RTGS, Cheque and NEFT together account for about 88% of all value from only 22% of transactions.
- UPI, IMPS and POS dominate everyday small-ticket activity.

Direction. Debits make up 67.3% of transactions (370,394) and ₹8.79 billion (53.5% of value). Credits are 32.7% of transactions (179,606) but ₹7.66 billion (46.5% of value), because the average credit (₹42,625) is much larger than the average debit (₹23,740).

Year-on-year. Activity is remarkably stable from 2019 to 2023, with 109,584 to 110,247 transactions and ₹3.25 to ₹3.34 billion in value each year. The peak was 2022 (110,247 transactions, ₹3.34 billion). The 2024 figures cover a single day (1 January, 295 transactions) and should not be compared with full years.

KYC status. 87.9% of transactions (483,457) are from Verified customers, 8.1% (44,288) from Pending and 4.0% (22,255) from Expired KYC. Together, Pending and Expired customers transacted about ₹2.03 billion.

7.3 Customer insights

Channels. The mobile app is the dominant channel at 37.9% of transactions (208,368) and reaches 74,103 of the 79,916 customers (92.7%). Web follows at 22.0%, then ATM (12.0%), POS terminal (11.1%), branch (10.0%) and API (7.0%). Success rates are almost identical across channels (91.9% to 92.1%).

Geography. Maharashtra, Karnataka and Tamil Nadu produce 45.0% of all transactions.

State	Transactions	Unique customers	Value (₹ billion)
Maharashtra	98,911	56,767	2.98
Karnataka	76,659	49,266	2.26
Tamil Nadu	71,747	47,406	2.15
Delhi	66,114	45,173	1.98
Gujarat	54,798	39,735	1.68
Telangana	49,610	36,933	1.43
West Bengal	43,950	33,831	1.32
Rajasthan	38,725	30,705	1.17

State	Transactions	Unique customers	Value (₹ billion)
UP	27,414	23,164	0.84
Punjab	22,072	19,241	0.65

Account types. Savings accounts generate 39.9% of transactions and 39.6% of value, followed by Current (25.0%), Salary (20.0%), NRI (8.0%) and Fixed Deposit (7.1%). Average account balances are similar across types (about ₹83,000 to ₹85,000).

7.4 Financial performance

Merchant categories. Retail (12.0%), Food & Dining (10.0%), E-Commerce (10.0%), Travel (8.0%) and Salary credits (8.0%) are the most active categories. Real Estate (3.0%), Fuel (4.0%) and Insurance (4.0%) are the smallest.

Loans. 192,288 transactions (35.0%) belong to customers with a loan.

Loan category	Transactions	Average EMI (₹)
Personal	47,810	6,789
Home	38,252	6,756
Auto	29,219	6,780
Business	28,863	6,764
Education	19,074	6,766
Gold	9,648	6,810
Unknown loan type	19,422	6,698
No loan	357,712	n/a

Personal and Home loans lead the loan book. About 10% of loan records (19,422) have a loan flag but no loan type recorded; these are reported as Unknown Loan Type.

7.5 Fraud and operational risk

- **Fraud value is concentrated.** RTGS alone accounts for ₹385.6 million, or 66% of all fraud value, followed by Cheque (₹106.0 million) and NEFT (₹69.2 million).
- **Fraud cases are mostly small.** UPI makes up the largest share of fraud cases (24.9%) because of its volume, but the average fraud transaction overall is ₹119,446, four times the average transaction, driven by RTGS and Cheque.
- **Fraud rate is stable over time** at 0.87% to 0.90% from 2019 to 2023.
- **Fraud rate varies little by channel or state**, ranging from 0.85% to 0.95% by channel and from 0.76% (Punjab) to 0.95% (UP) by state.
- **KYC status and credit score do not separate fraud risk.** Fraud rates are 0.84% to 0.89% across KYC statuses and 0.87% to 0.92% across credit-score bands.
- **Operational failures.** 21,905 transactions failed (₹697 million) and 11,019 were reversed (₹326 million). A further 10,829 (₹334 million) are pending.

8. Insights and Recommendations

1. **Prioritise RTGS and Cheque for fraud controls.** Add enhanced verification, velocity checks or manual review for high-value RTGS and Cheque payments, where the fraud rate is 1.3 to 2.1 times the average and fraud losses are concentrated.
2. **Close the KYC gap.** Run a re-verification campaign for the customers behind the 66,543 Pending and Expired KYC transactions and consider transaction limits until KYC is refreshed.
3. **Protect UPI and mobile reliability.** UPI and the mobile app carry the most volume and the largest customer reach, so uptime and failure monitoring here protects the most customers.
4. **Reduce failed and reversed payments.** A 6% combined failure and reversal rate is worth about ₹1.0 billion. Root-cause analysis by transaction type and channel would identify quick wins.
5. **Grow outside the top three states.** Maharashtra, Karnataka and Tamil Nadu dominate. Punjab, UP and Rajasthan are under-penetrated and are candidates for targeted acquisition.
6. **Improve loan data capture.** Make Loan_Type mandatory when a loan flag is set so that the Unknown Loan Type group disappears.
7. **Cross-sell by account type.** Savings and Current customers are the largest bases for loans, investments and insurance offers.

9. Data Quality Notes and Limitations

- **Highly uniform patterns.** Transactions are spread almost perfectly evenly across hours, years, months and fraud rates by segment. This suggests the dataset may be simulated, so the findings are best treated as a demonstration of analytical method rather than a description of a real institution.
- **Partial final year.** 2024 contains only 295 records from 1 January and distorts any year-over-year comparison that includes it.
- **Extreme values.** Transaction amounts are highly skewed (mean ₹29,907 vs median ₹2,036). 1,746 transactions are at or above ₹1 million and the maximum is ₹10 million, so averages should be read alongside medians.
- **Total_customers counts rows, not people.** The measure counts customer ID occurrences (550,000), so it behaves as a transaction count. Charts that need a head count should use Unique_Customers.
- **Unique customers do not sum across categories.** A customer can hold several account types and use several channels, so segment totals exceed 79,916.
- **Chart titling.** The Customer Insights and Financial Performance pages both contain a chart titled "Total Customers by Channel"; checking that titles match their fields is a useful final polish.
- **Currency.** Amounts are shown in rupees (₹), based on the Indian states, UPI, IMPS, NEFT and RTGS payment types in the data. Please confirm this if the source documentation says otherwise.
- **Fraud flag.** Fraud analysis relies on the Is_Fraud label in the source data; no independent fraud model was built.

10. Conclusion

The Banking Transaction Analytics dashboard consolidates 550,000 transactions into three focused views that answer who is transacting, how, and where the risk lies. The analysis shows a stable, mobile-led business with a 92% success rate and low overall fraud (0.89%), but with high-value risk concentrated in RTGS and Cheque, a KYC compliance gap affecting 12% of transactions, and uneven geographic reach. The standard KPI measures and slicers make the dashboard easy to extend and reuse.

11. Future Enhancements

- Add a date table and time-intelligence measures (year-on-year and month-on-month growth).
- Add fraud analysis by hour, amount band and customer segment, and a drill-through page for fraud cases.
- Add average and median transaction amount, and failed and reversed rate measures.
- Add a credit-risk view combining credit score, EMI and account balance.
- Schedule data refresh through a Power BI gateway and publish to the Power BI Service with row-level security.

Appendix: Tools and Technologies

Excel and Python (initial data analysis), MySQL (SQL queries for business problems), Microsoft Power BI Desktop (Power Query, DAX, report visuals).